
Jess Sullivan

Full Stack Engineer · DevSecOps · Computer Vision · Kernel & Security Research · ML/HPC

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Languages

Python
TypeScript, Rust
Zig, Chapel, Haskell
Go, C++, R, Shell

ML/AI & HPC

Fine-grained classification
TensorFlow, pandas
NUMA-aware horizontal parallelism
Realtime model evaluation & SLM inference

Systems & Security

GhidraScript
Frida, ILSpy
Enterprise MFA systems & standards
OpenTelemetry

Infrastructure

Nix, Bazel
OpenTofu, Ansible
RKE2 (k8s)
GitLab AutoDevOps

Experience

Systems Analyst (DevSecOps) @ Bates College

(2024–Present)

Scalable enterprise systems supporting staff, faculty, and ILS team. Legacy modernization; bespoke Ansible extensions/roles/plugins; 24/7 CVE mitigation; SAML and application interoperability; OpenTelemetry reporting; CI/CD pipelines (GitLab AutoDevOps, OpenTofu, RKE2 + Rancher); leading IaC adoption college-wide.

Noteworthy projects:

- Built a property-based orchestrator and instrumentation harness for 1980s-era degree-management C in [Haskell](#) + [Python](#) (QuickCheck, Cabal, podman-compose for dev parity, FPM for packaging, AutoDevOps for CI/CD); turned an unautomatable legacy codebase into a verifiable, traceable, k8s-friendly workload.
- Overhauled and automated the event-management lifecycle ([C#](#), [Go](#), [Ansible](#)).
- Led adoption of horizontally-scalable [Apache Solr](#) instances across multiple public and private indexing and search applications.
- Built out internal ACME-first certificate-management and DNS libraries, templates, and tooling.
- Drove enterprise secret-management practice and authentication / authorization college-wide; developed multiple SAML, LTI, Shibboleth, and bespoke TOTP / OAuth integrations; led adoption of [KeePassXC](#), [firewalld](#), and [fail2ban](#) inside declarative Ansible workflows; OTEL / LGTM / Tempo telemetry stack in production.

Fabrication Laboratory Manager @ Cornell CALS

(2021–2022)

Developed and taught rapid fabrication curricula. [OpenSCAD](#), [Fusion 360](#), [Inkscape](#), [C++](#). GitHub/Linear project management.

Computer Vision Software Engineer @ Macaulay Library, Cornell Lab of Ornithology

(2018–2022)

Developed & launched [Merlin Sound ID](#), a production fine-grained audio classification system now used by millions worldwide. Led R&D on internal ML annotation tooling, model evaluation APIs, and end-to-end MLOps pipelines. Built real-time inference demos deployed at scale.

Stack:

- **Model Training:** Python ([TensorFlow](#), [NumPy](#), [Pandas](#), [Matplotlib](#), [Jupyter](#))
- **Web & Inference:** [Flask](#), [TypeScript](#), [React](#), [Vue](#), [Docker](#), [WebAssembly](#), [React Native](#), [Swift](#)
- **Infrastructure:** EC2, Heroku, BitBucket CI/CD, production model deployment at scale

Full Stack Contracting and FOSS

(Ongoing)

Ongoing contributor and community member of numerous open source projects including—and not limited to—the [Apache Foundation](#), [rspamd](#), [Chapel-lang](#), [numtide/nix-vm-test](#), [manaflo-ai/cmux](#), [diku-dk/Futhark](#), [xCaddy](#), [libdns](#), Skeleton UI, [Klipper](#), [Joplin](#), [FFT.js](#), [KeePassXC](#), [svelte-superforms](#), [ShikiJS](#), [ggplot2](#), the [Rocky Enterprise Linux Foundation](#), [Liqo](#), [Budgie DE](#), [Mason](#), along with authoring numerous FOSS automation tools, libraries and open source utilities. Expanded client list and customer references available upon request.

- Startup work: **Dover Micro** (2017), **Adaptive Motorsport** (2018); long thread of personal HID, XR, and sensor-fusion projects.
- Web GIS tools for **National Park Service**, **Foundation for Healthy Communities**, **GPRED**, **Northern Border Regional Commission**. Presented at 2019 AAG Annual Meeting in Washington, DC (§3).

- Fine-grained image classification with **MushroomObserver.org** and **Visipedia**; early adopter of CNN-based species ID at scale.

Current stack & technologies:

- **Web & product:** SvelteKit Runes, Bun, Vite 8/Rolldown, TypeScript (TS7), Tailwind CSS, Skeleton UI, Effect TS, Postgres, FingerprintJS; auth, scheduling, mapping, and site systems.
- **Operations & observability:** Tempo/Grafana, Caddy, Tailscale Operator, on-prem RKE2, Podman Compose.
- **Infrastructure:** Nix flakes, Bazel RBE, justfile, OpenTofu, sops+age, Ansible.

Research:

- **Reverse Engineering & Binary Analysis:** GhidraScript, Frida, ILSpy, Mitre Caldera, Zig. Firmware RE and NVMe XRAM recovery (see §3).
- **Author of numerous Zig capability libraries** with C ABI surfaces: [zig-crypto \(docs\)](#), [zig-notify \(docs\)](#), [zig-keychain \(docs\)](#), [zig-ctap2 \(docs\)](#).
- **linux-xr** — Rocky Linux 10 RPM kernel lane carrying XR display patches and [Dirty Frag security backports](#). Backported **CVE-2026-31431**, **CVE-2026-43284**, and **CVE-2026-43500** into 6.1.y ahead of public disclosure.
- **Functional & Heterogeneous Compute:** Futhark/WebGPU, ESDT monads, fine-grained classification systems, Rust SIMD, Nix build systems, deeper WASM integration, and WASM-native inference pipelines.

Volunteer & Community

First Fellow @ D&M Makerspace, Plymouth State University

(2017–2020)

Taught **Advanced GIS Programming & Intro to Electromechanics**. COVID-19 response: coordinated regional makerspace network for medical PPE manufacturing.

Membership Chair & 3D Printing Captain @ Ithaca Generator

(2020–2022)

Led 501(c)(3) makerspace through rapid growth; coached hundreds via “Fusion 360 for 3D printing” series.

Publications

Reitsma, L.R., Burns, C., & **Sullivan, J.** (2019). *Poecile atricapillus* (Black-capped Chickadee) Feeding *Catharus guttatus* (Hermit Thrush) Nestlings. *Northeastern Naturalist*, 26(2). [doi:10.1656/045.026.0213](#)

Sullivan, J. (2026). Recovering Write-Protected NVMe SSDs Through USB Bridge XRAM Injection: Bypassing the ASMedia ASM2362 Firmware Opcode Whitelist. [recovery-paper.pdf](#)

Presentations:

Sullivan, J. (2019). Web GIS: Telling Stories & Solving Problems. *Association of American Geographers Annual Meeting*, Washington, DC.
